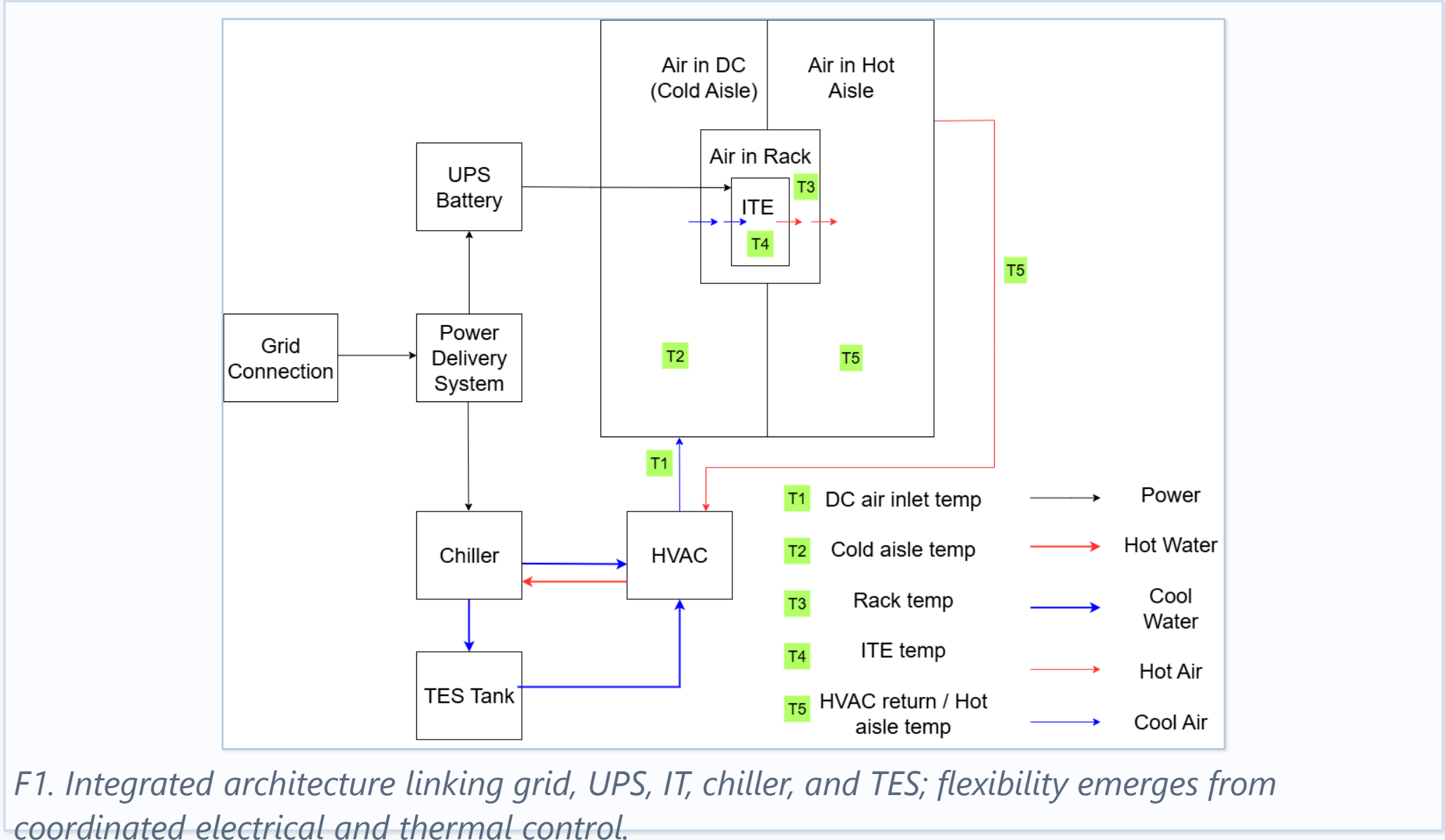


Characterisation and Quantification of Data Centre Flexibility for Power System Support

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Core message: an integrated IT-UPS-cooling strategy can reduce operating cost and provide duration-certified flexibility to the grid.



Problem and Motivation

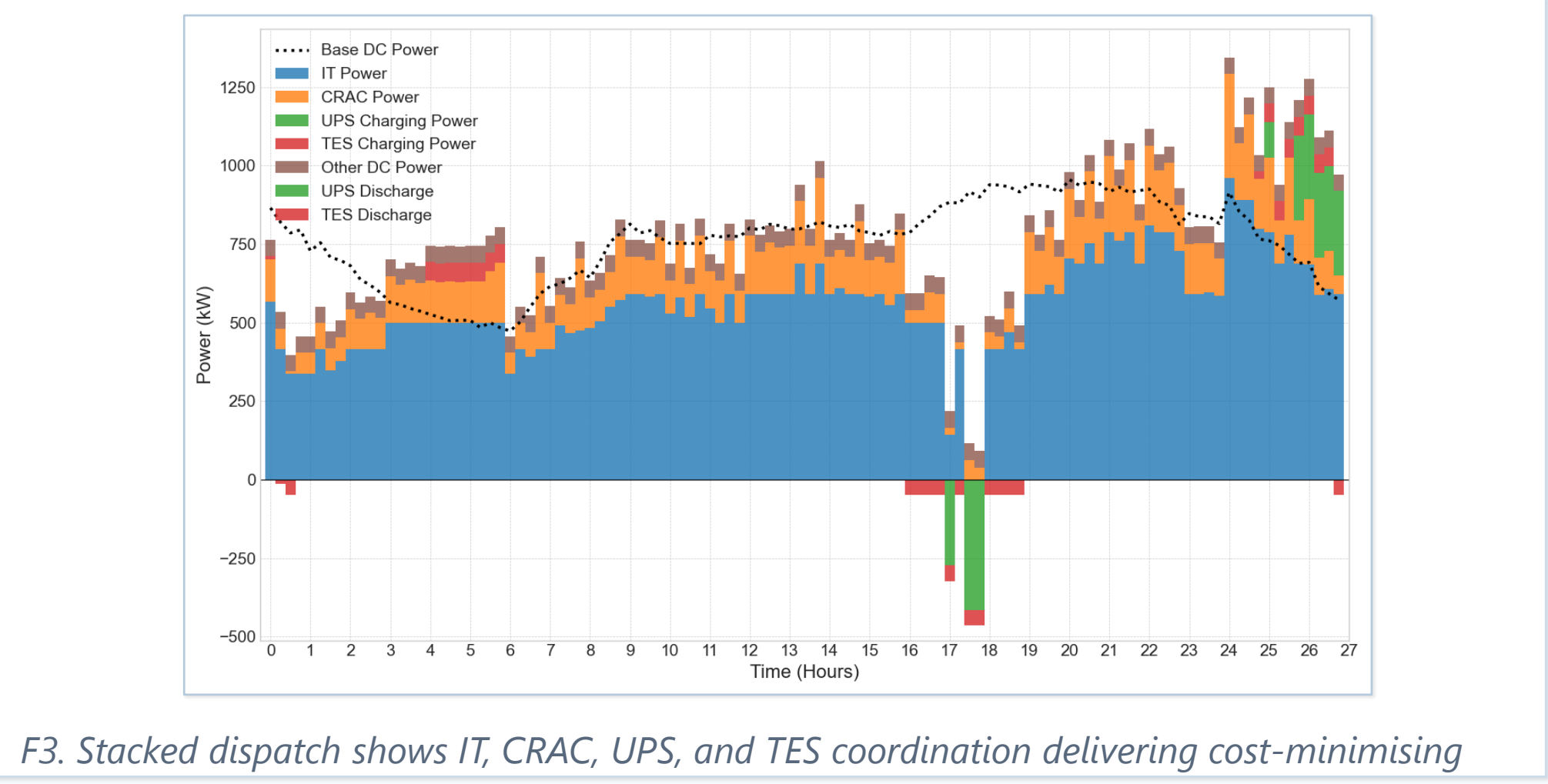
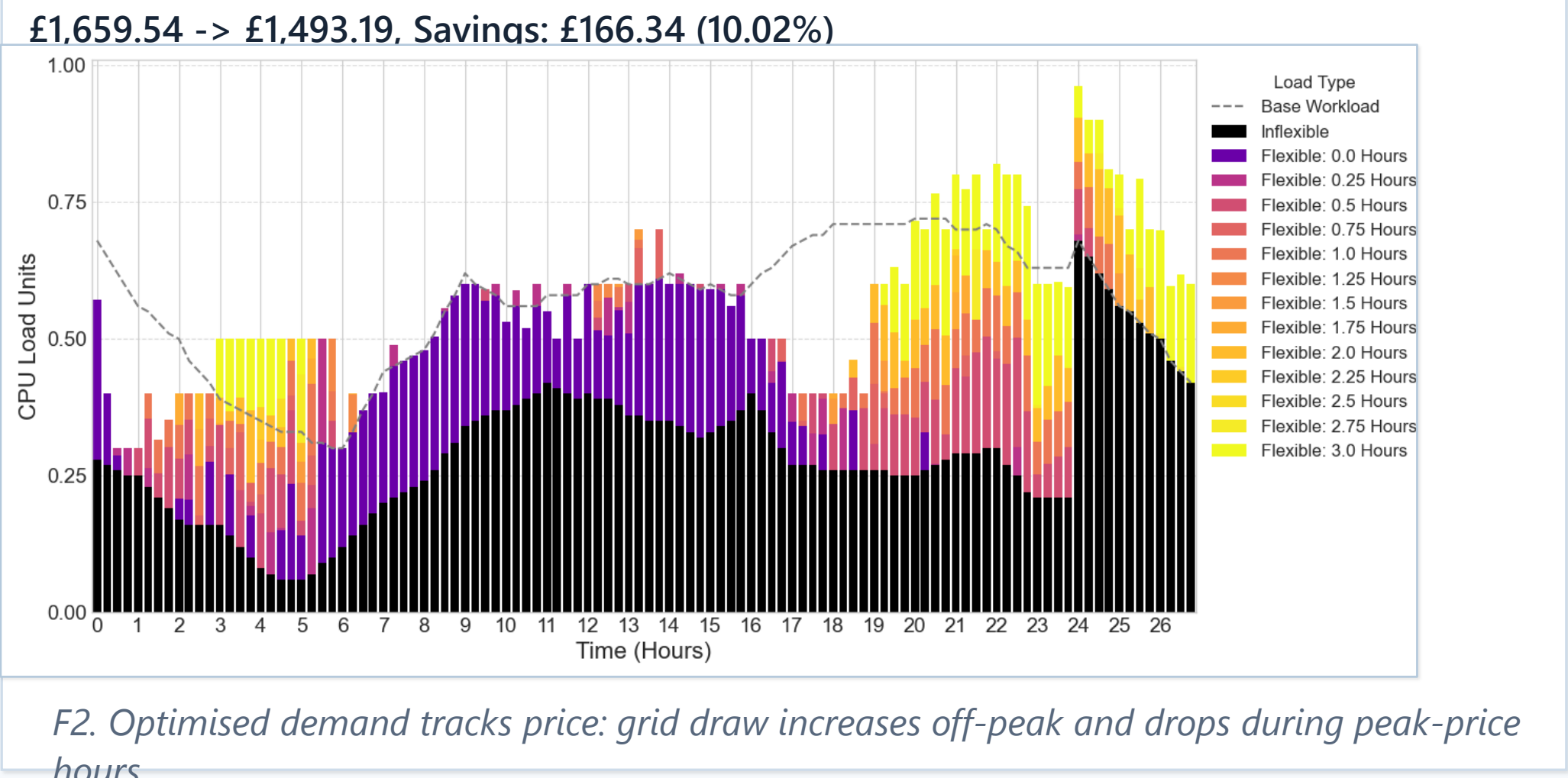
Global digitalisation and AI are accelerating data-centre electricity demand while power systems integrate variable renewables and new electric loads. This increases peak stress, ramping pressure, and local network congestion. Data centres therefore have a dual role: they are part of the flexibility challenge, but they also host controllable assets that can support balancing when operated as an integrated system.

Research Objective

This work develops and tests a whole-facility framework with two contributions. First, it computes a least-cost day-ahead operating baseline by co-optimising IT scheduling, UPS operation, and cooling/TES dispatch. Second, it quantifies duration-certified flexibility: for any requested power deviation and start time, it computes the maximum feasible provision duration.

Integrated Model

Flexibility is represented through three coupled subsystems. IT flexibility comes from shifting delay-tolerant workload while preserving constant computational demand and respecting tranche delay limits. UPS flexibility comes from bounded charge/discharge power, state-of-charge limits, and non-simultaneous charging/discharging constraints. Cooling flexibility comes from chiller modulation, TES charge/discharge, and thermodynamic temperature dynamics for inlet air, cold aisle, racks, IT components, and hot aisle. The model uses 15-minute time steps over a 24-hour horizon plus a 3-hour extension window. A 1 MW IT facility is analysed with UPS capacity of 600 kWh and TES capacity of 1000 kWh, using day-ahead electricity prices.

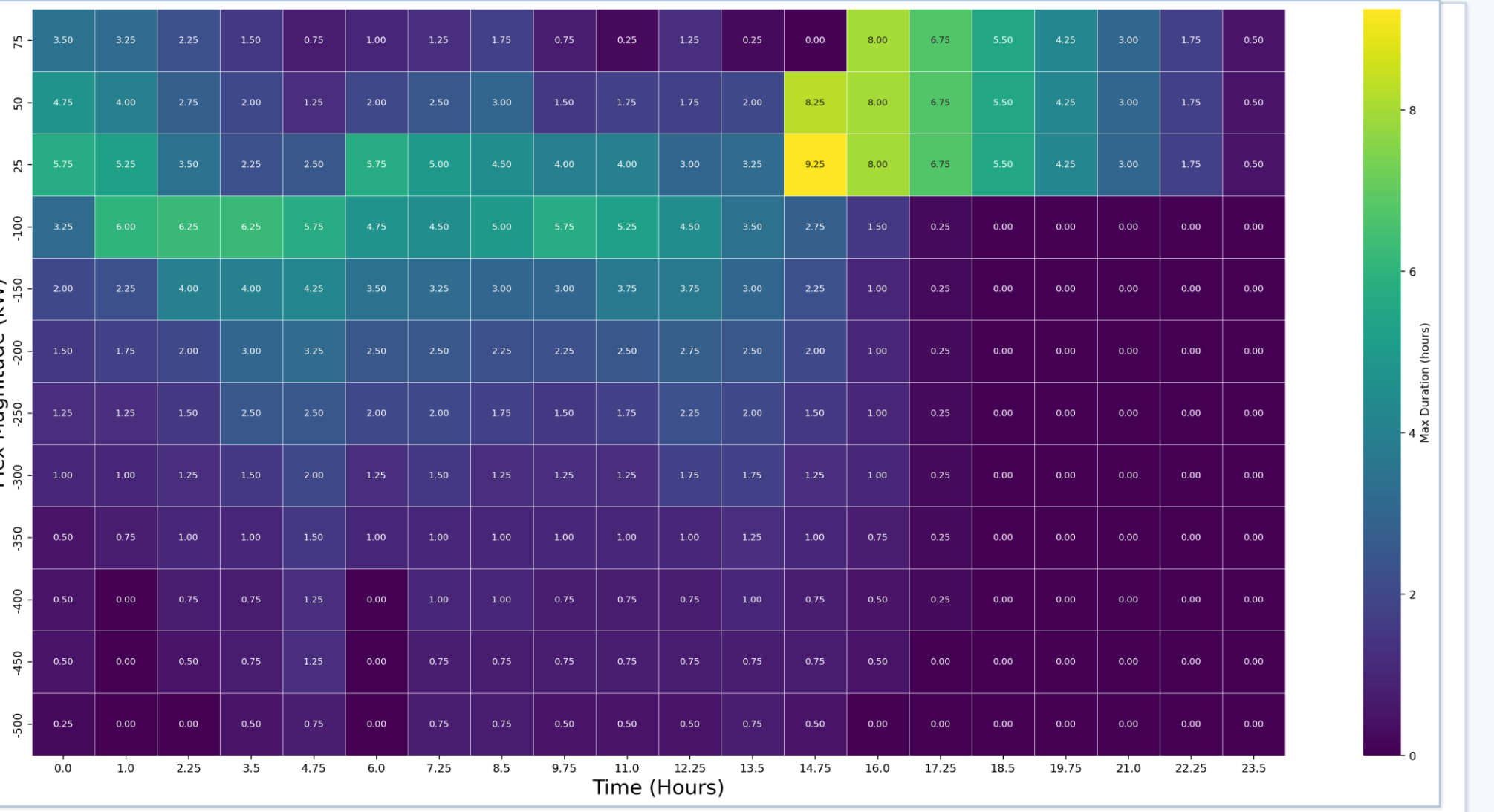
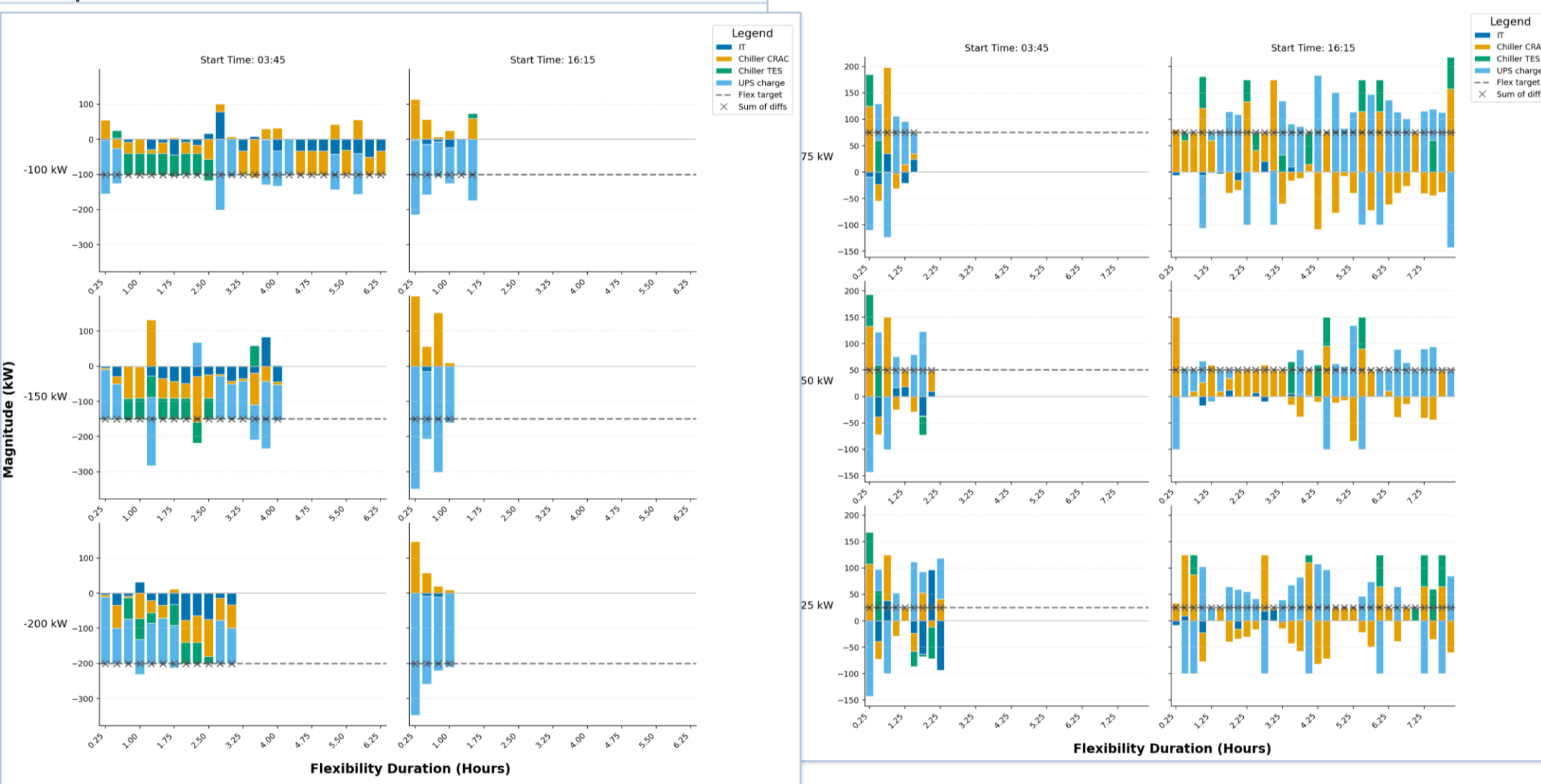


Scenario Design

Scenario 1 establishes a no-flexibility base-case cost benchmark. Scenario 2 co-optimises workload, UPS, and cooling/TES to minimise operating cost under physical constraints. Scenario 3 performs a duration-aware flexibility test on top of the Scenario 2 baseline. For each start time t_0 and requested deviation ΔP , the model searches for the maximum feasible duration τ while enforcing thermal and operational constraints and a 12-slot recovery window so post-event operation remains feasible.

Key Results (Cost and Dispatch)

Cost optimisation reduces daily operating cost from £1,659.54 to £1,493.19, delivering £166.34 savings (10.02%) without additional hardware investment. During low-price periods, the site increases grid draw to pre-charge TES/UPS and process more IT load; during peak-price periods, it reduces grid draw through deferred IT execution, UPS discharge, and cooling reallocation.



Key Results (Flexibility Asymmetry)

The flexibility heatmap reveals strong temporal structure and clear asymmetry. Upward flexibility (negative ΔP) is strongest when IT deferral headroom is available and is reinforced by cooling reduction. Downward flexibility (positive ΔP) is concentrated later in the day and relies more on CRAC/TES and UPS charging because advancing IT workloads is constrained without pre-conditioning. A representative contrast is -100 kW upward flexibility for 6.8 h at 00:15 versus 0.2 h at 17:30. Asset-level panels show coordinated internal counter-movements that preserve the required net response.

Practical Implications

The framework turns abstract flexibility potential into market-relevant, duration-certified products for reserve, balancing, and price-responsive demand. Operators can optimise cost first, then stack flexibility services using residual capability. System planners gain time- and magnitude-specific estimates of dependable demand-side response from data-centre assets. Because duration is certified rather than assumed, outputs can be mapped directly to products with explicit start-time and magnitude commitments.

Limitations and Next Work

Future work: explicit revenue stacking, pre-conditioning windows for more symmetric downward IT participation, and multi-site portfolio aggregation.

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